

Exercise Physiology

Contents

- ventilation in response to exercise

 - gas exchange

 - factors on ventilation

(chemical, non-chemical, neurogenic, temperature and combined factors)

- ventilatory equivalent for oxygen

- effects of lactate on ventilation

- changes in blood pressure during exercise

- changes in heart rate and stroke volume during exercise

- cardiac output distribution during exercise

Exercise physiology

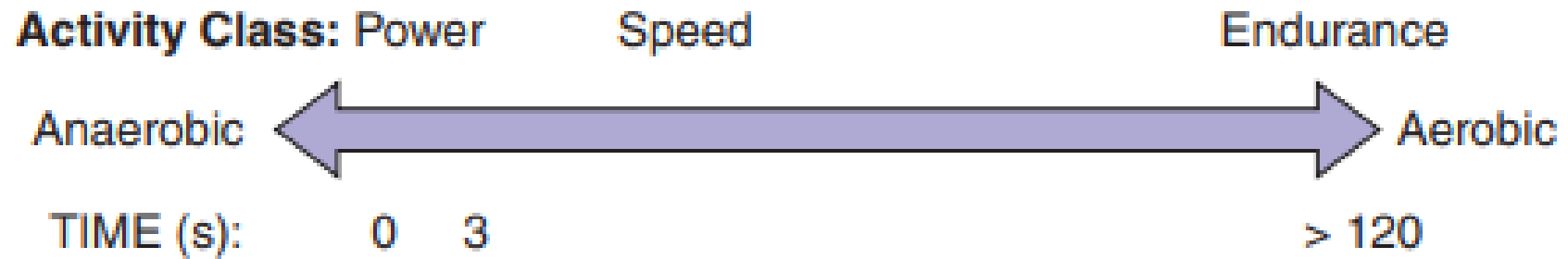
The study of

- body function under acute and chronic exercise
- body response to intense demand and changes during exercise

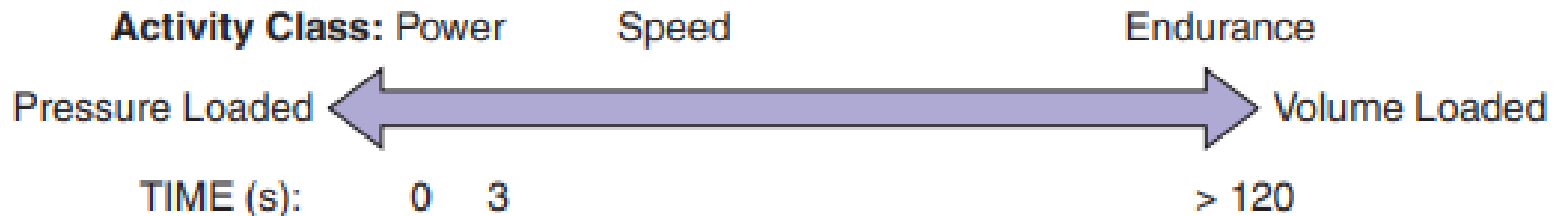
Acute exercise-the performance of a single bout of exercise

Chronic exercise-exercise in a length of time which changes take place in multiple physiological systems during exercise training

Physical activities as energetic events



Physical activities classified on a hemodynamic continuum



The cardiorespiratory system

The pulmonary system

- to provide adequate supply oxygen
- absorbs oxygen from ambient.....into the blood

The cardiovascular system

- to carry oxygen to working muscle

Heart and lung coordinate precisely

- to match the increased metabolic demand for oxygen
- to maintain O_2 and CO_2 homeostasis

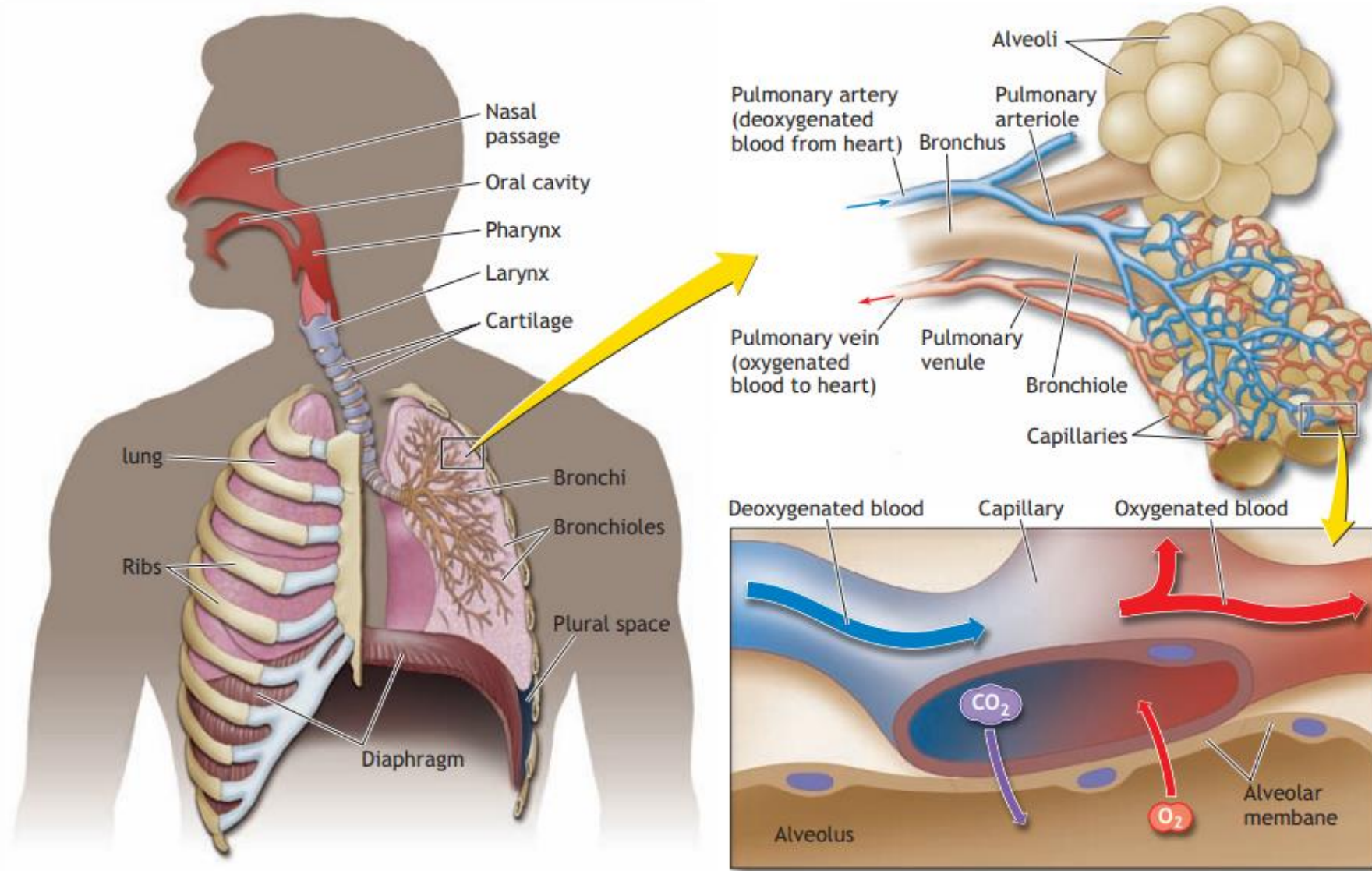
Integration of the pulmonary and cardiovascular systems for oxygen

-delivery

-extraction

-utilization (consumption)

$$\begin{array}{ccc} \text{Oxygen Consumption} = \text{Cardiac Output} & & \text{Arteriovenous Oxygen Difference} \\ \text{(Utilization)} = \text{(Amount Delivered)} & \times & \text{(Amount Extracted)} \\ \text{Central Factor (Heart)} & & \text{Peripheral Factor (Tissues)} \end{array}$$

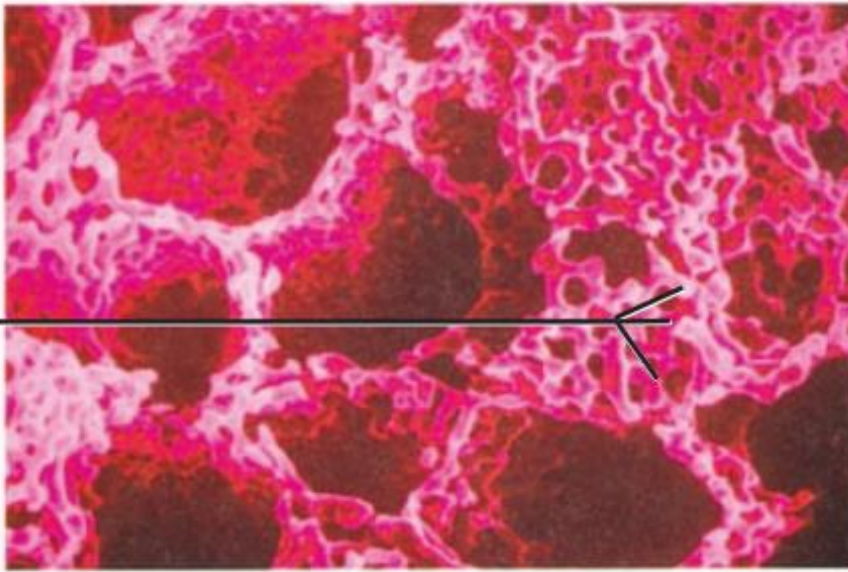


Pulmonary structures and gas exchange function

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Lung capillary bed

A



Alveolar air

Alveolar
capillaries

Respiratory
membrane
separating
the alveolar
air and
blood

Red blood
cell

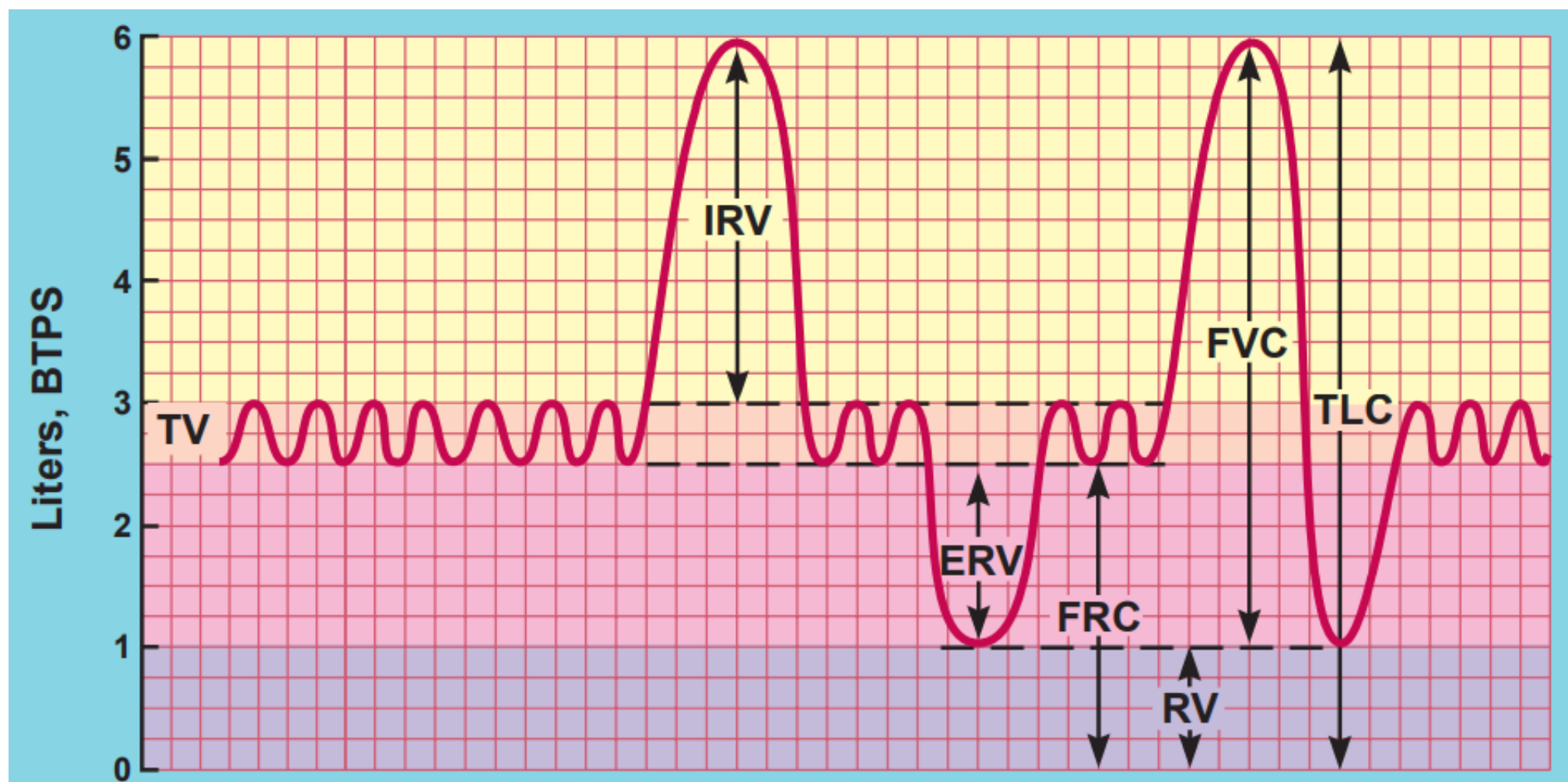
B



Lung capillaries

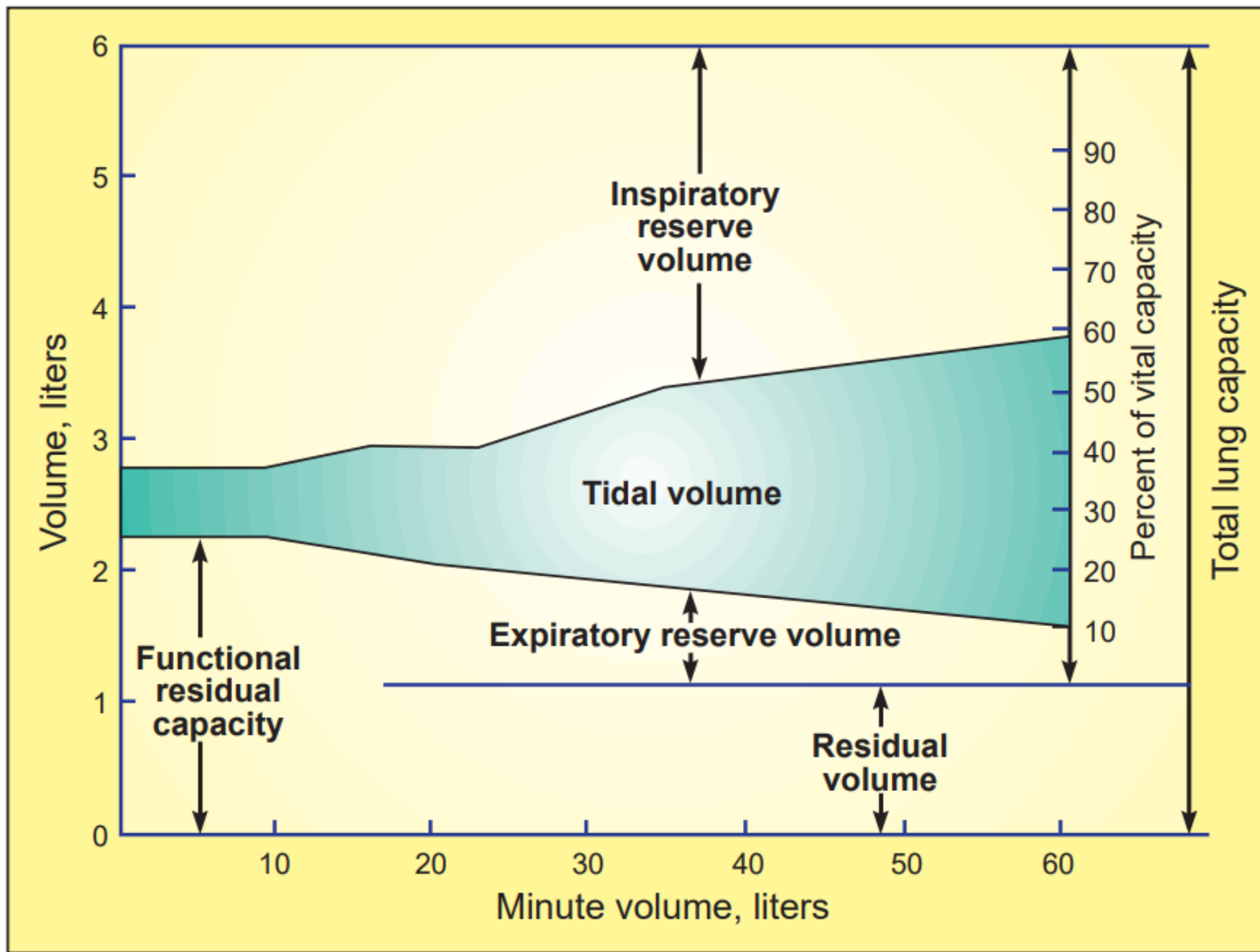
- very thin membrane separating
- alveolar air
- red blood cells

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Static measures of lung volume and capacities

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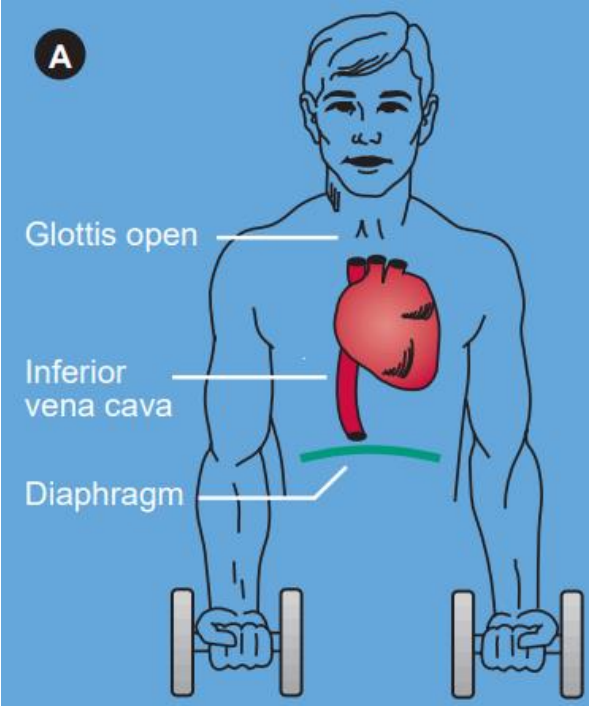


Lung volumes during rest and exercise

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Increased in minute ventilation during exercise

- $\uparrow$ TV..... $\uparrow$ breathing rate
- adjusted unconsciously
- modify breathing does not benefit exercise performance
- natural breathing reduces the energy cost



Valsalva maneuver

- closing glottis...max. exhale.....increased intra-thoracic pressure
- blood pressure briefly increase
- inferior vena cava compress
- reduced venous return.....decreased arterial BP

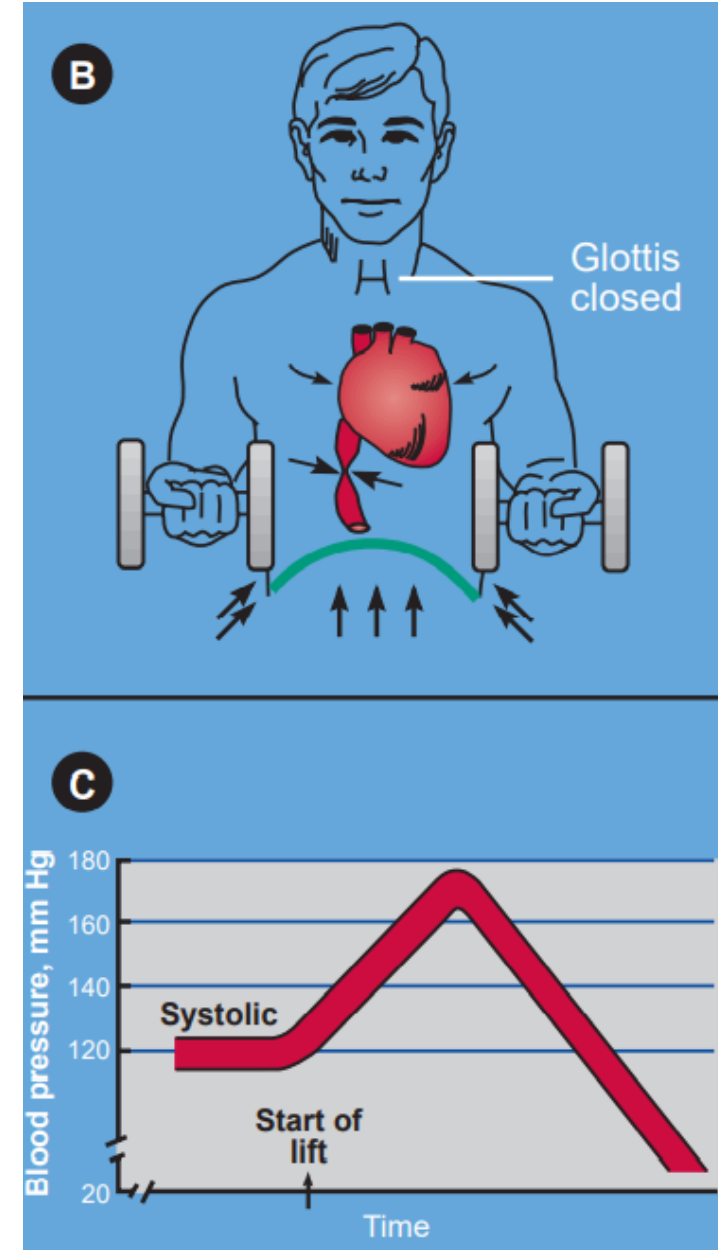
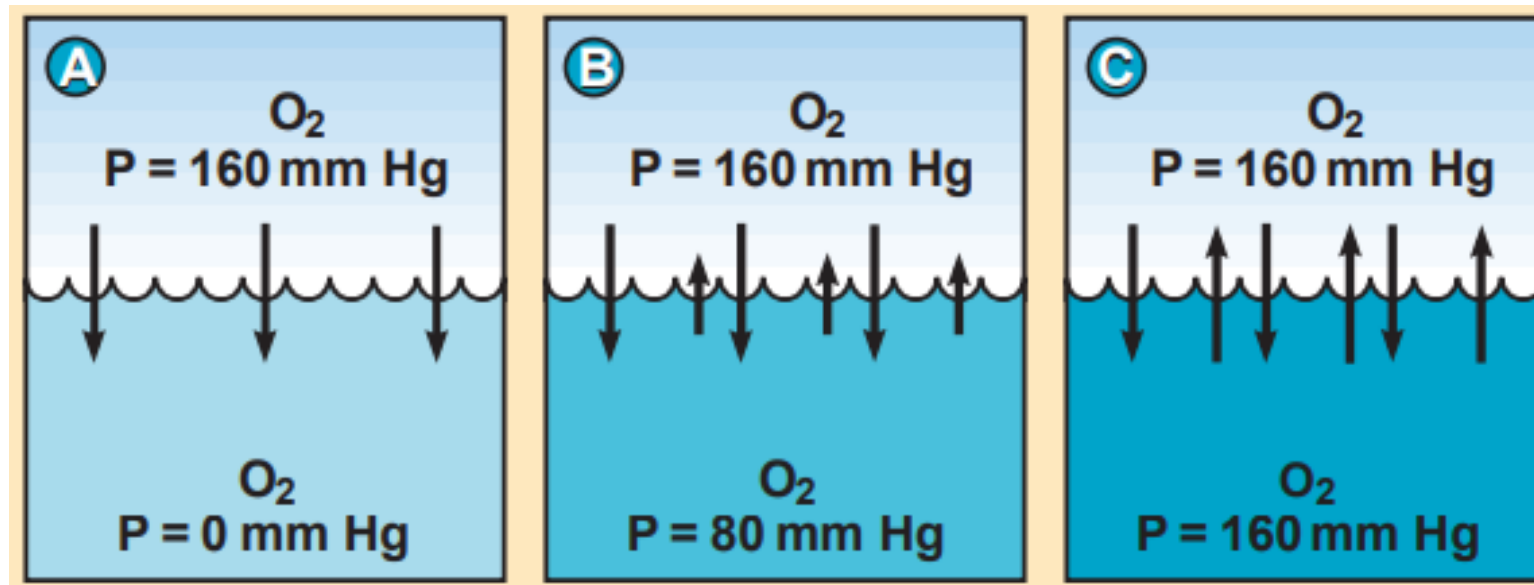


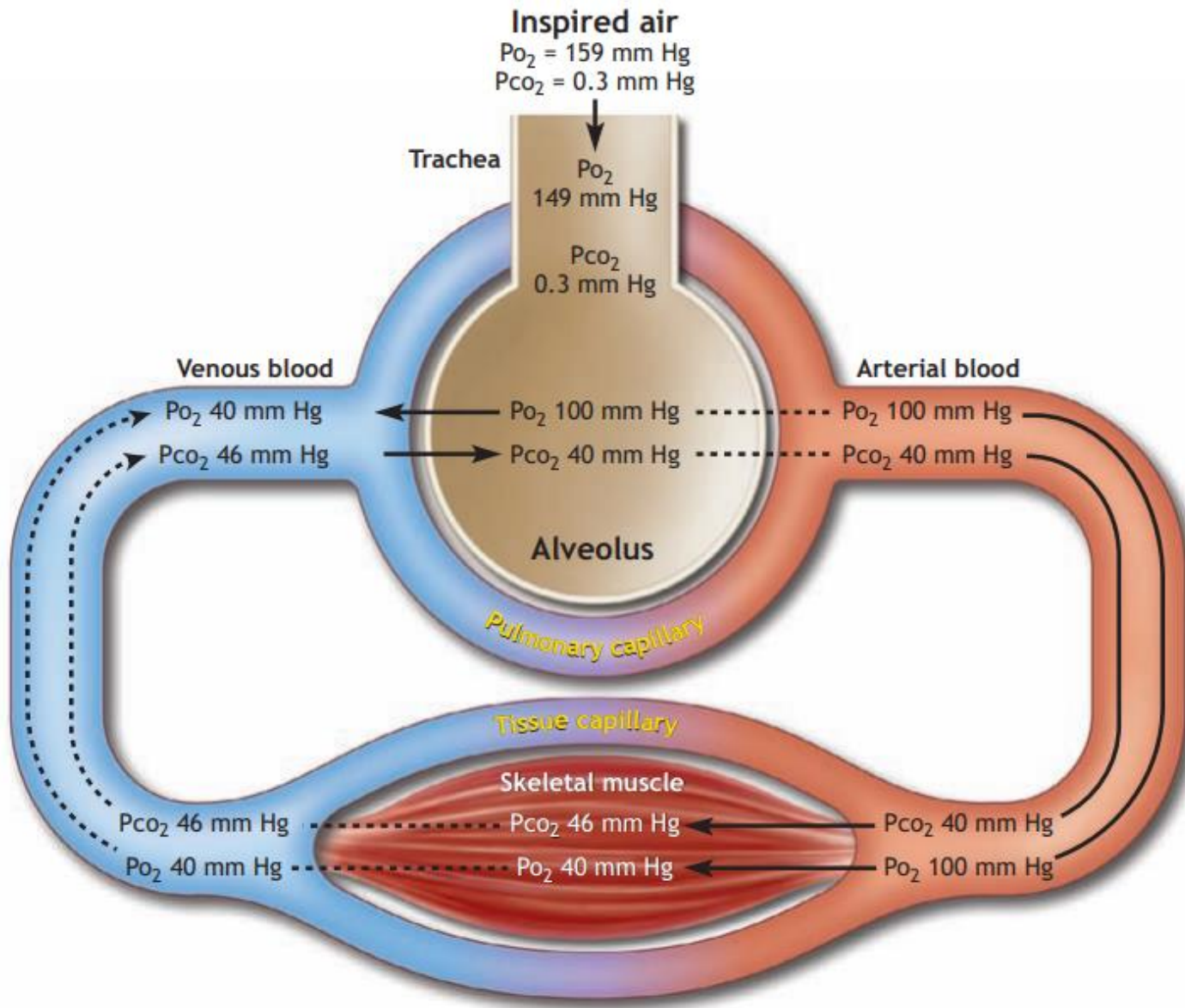
Table 9.2**Percentages, Partial Pressures, and Volumes of Gases in 1 L of Dry Ambient Air at Sea Level**

GAS	PERCENTAGE	PARTIAL PRESSURE (at 760 mm Hg)	VOLUME OF GAS (mL·L ⁻¹)
Oxygen	20.93	159 mm Hg	209.3
Carbon dioxide	0.03	0.2 mm Hg	0.4
Nitrogen	79.04 ^a	600 mm Hg	790.3



Partial pressures and solution of oxygen in water

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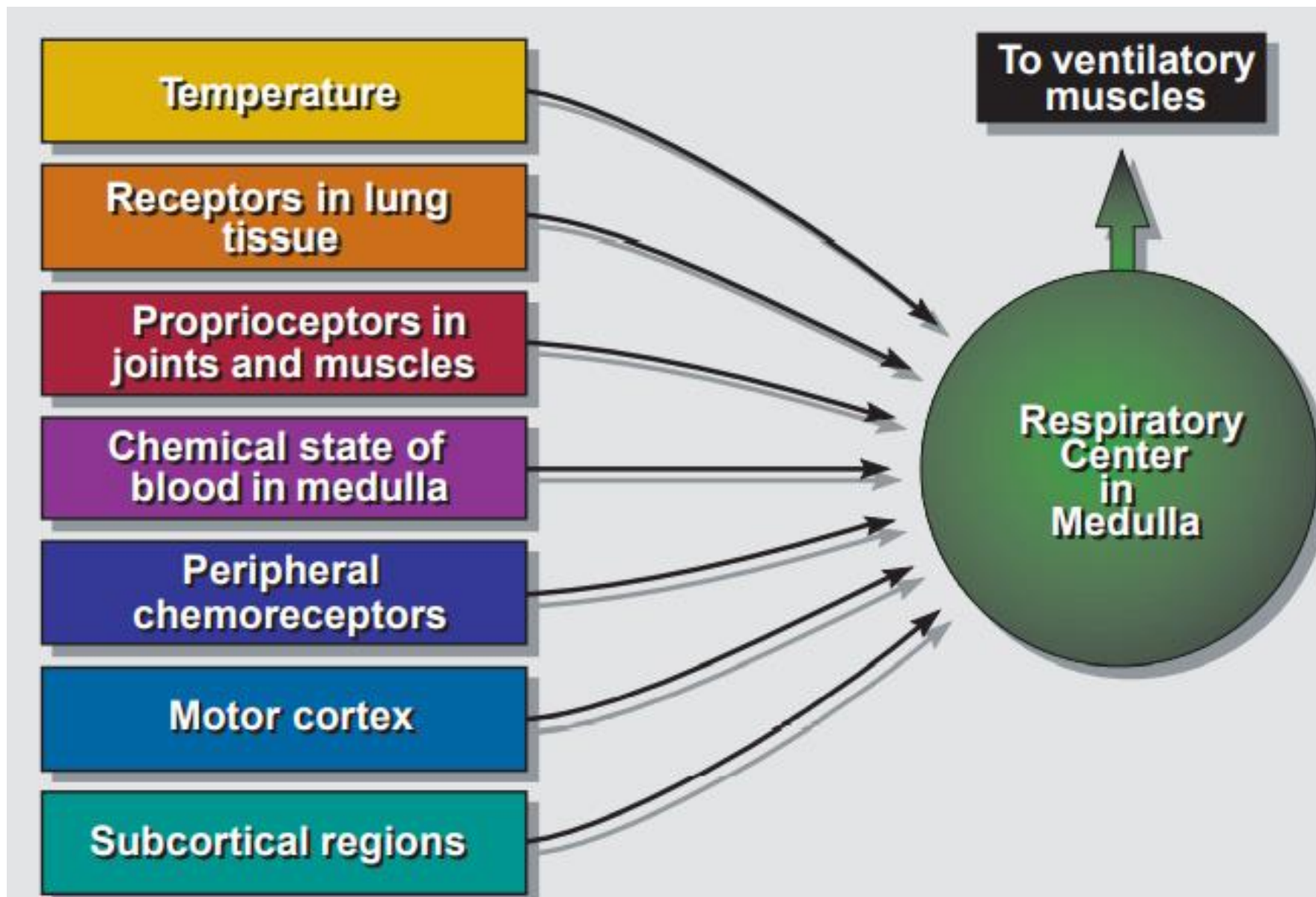


Gas transfer at rest

-Partial pressures in alveola and arterial blood are equal

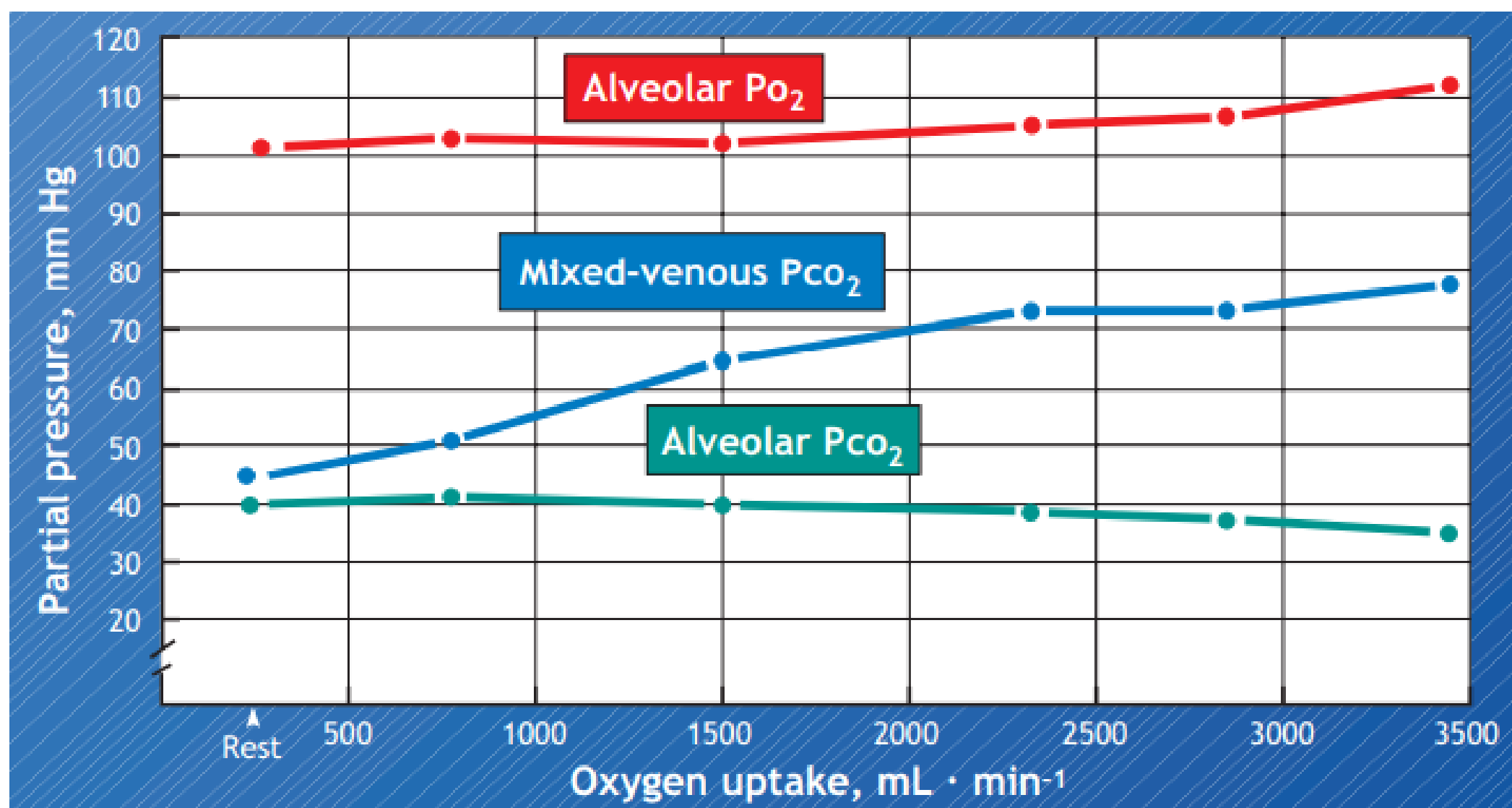
-extraction rate depends on
Diff. between pressures in alv/art

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Medullary control in response to factors

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Chemical factor on ventilation during exercise

- chemical not fully explain the increased ventilation

- arterial P_{O_2} , P_{CO_2} and acid do not increase ventilation as much as during heavy exercise

- Increased CO_2 and $[H^+]$ during strenuous exercise..... $\uparrow\uparrow$ ventilation...
..... $\downarrow\downarrow\downarrow P_{CO_2}$ $\uparrow\uparrow O_2$

Non chemical factors of ventilation during exercise

- ventilation increases rapidly almost at the first ventilation cycle
- followed by 20-s plateau
- further increased in ↑ ventilation... . to match demand of metabolic gas exchange
-
-
- when exercise stops...
- ...
- ..ventilation decreases to 40% of final exercise....
- ...slowly returns to resting levels

Neurogenic factor of ventilation during exercise

- cortical influence from motor cortex

 - in anticipation

 - during exercise

 - cortical outflow abruptly increases ventilation when exercise begins

- peripheral influence

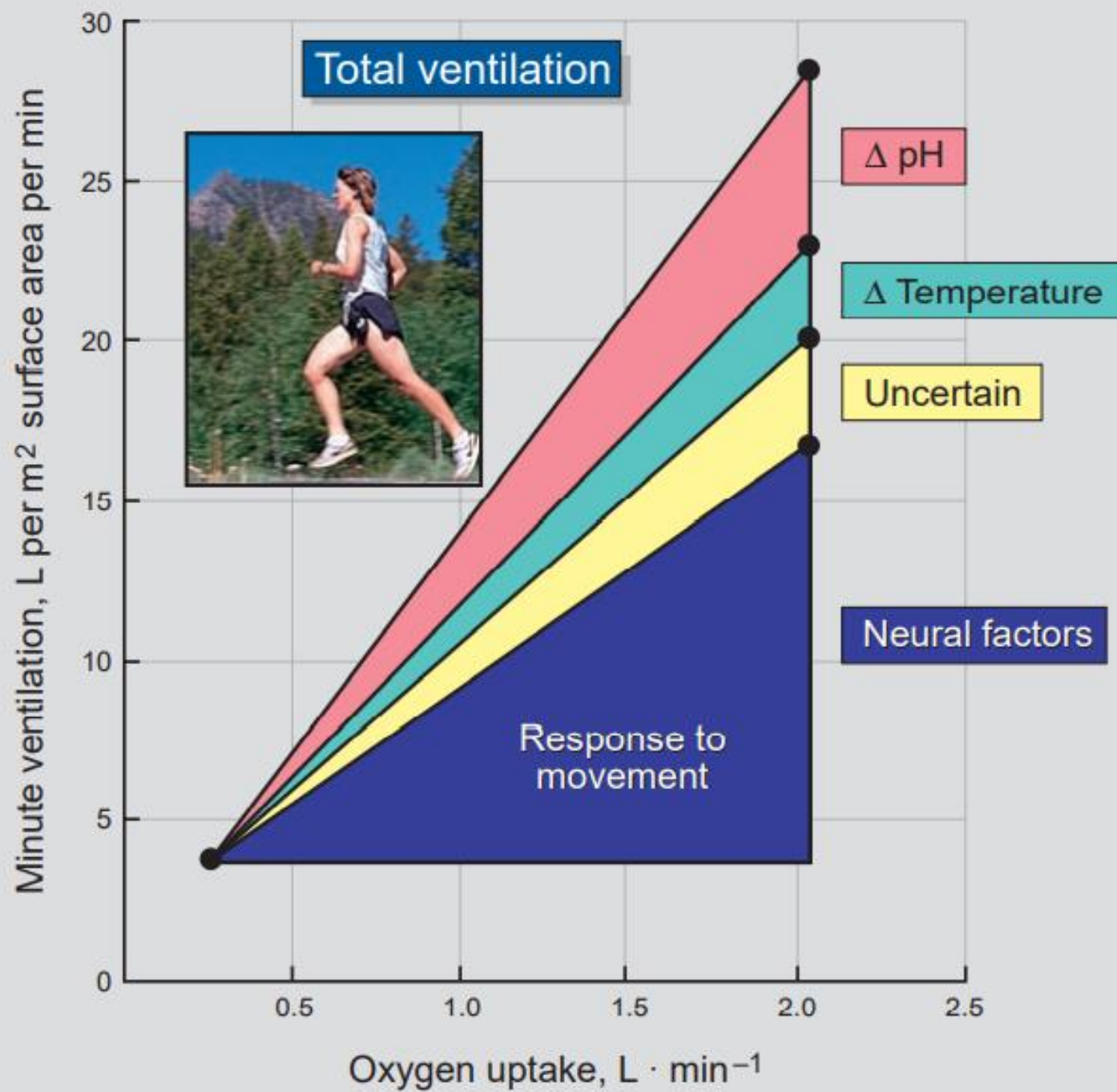
 - sensory from joints, tendons, muscles (mechanoreceptors)

Influence of temperature

- however, rapid change in ventilation is not explained by slowly increased body temp.

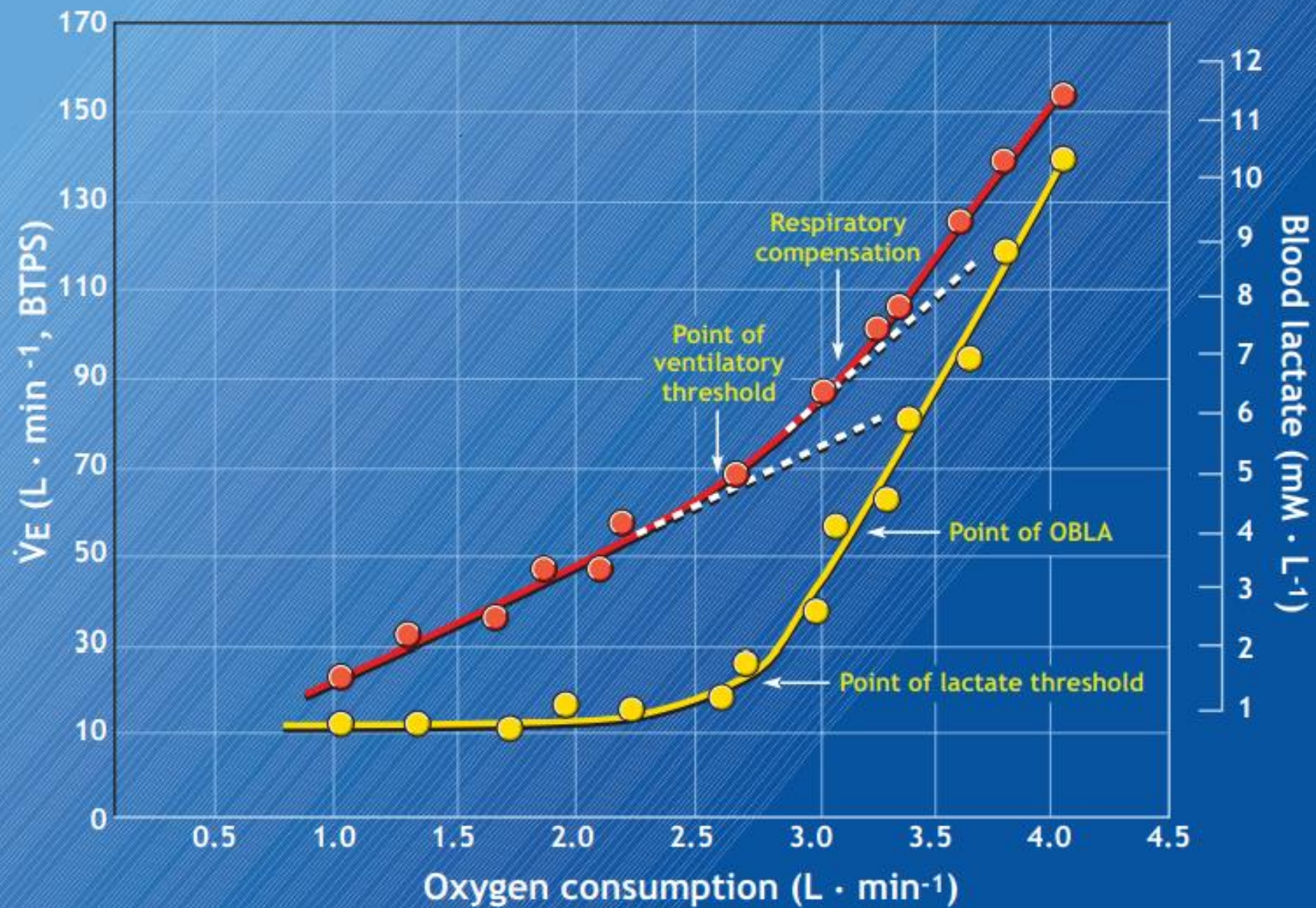
Integrated regulation

- combined effects of several chemical and neural stimuli



Ventilatory equivalent for oxygen in steady-Rate exercise

- $\dot{V}_E/\dot{V}O_2$ represent ratio of minute ventilation to oxygen uptake
- in healthy young adult is about 25
(25 L air breath per L oxygen consumed)
- up to 32 in 6-year-old children
- oxygen supply and utilization satisfy energy needed by muscles
- lactate does not exceed its removal



Minute ventilation Blood lactate

Ventilatory equivalent for oxygen in **Non**-Steady-Rate exercise

Ventilatory threshold in graded exercise

- disproportion between minute ventilation to oxygen uptake
- ventilatory equivalent increases to as high as 35 or 40 in max exercise
- no longer linked with oxygen demand but rather carbon dioxide instead (from buffering of lactate due to anaerobic metabolism)
- or anaerobic threshold



Ventilatory equivalent for oxygen in **Non**-Steady-Rate exercise (continued)

- excess CO₂.....↑↑pulmonary ventilation
- further disproportionately increases VE/VO₂
- the onset of anaerobiosis

Ventilatory equivalent for oxygen in **Non**-Steady-Rate exercise (continued)

Onset of blood lactate accumulation

- when blood lactate increases above baseline level of 4 mM/L

- the point of onset of blood lactate accumulation (OBLA)

(normally from 55% to 65% of VO_2max . in healthy untrained subjects

=80% VO_2max in highly trained athletes

Causes of OBLA unclear

- hypoxia. However, lactate can form even with enough oxygen in muscle.

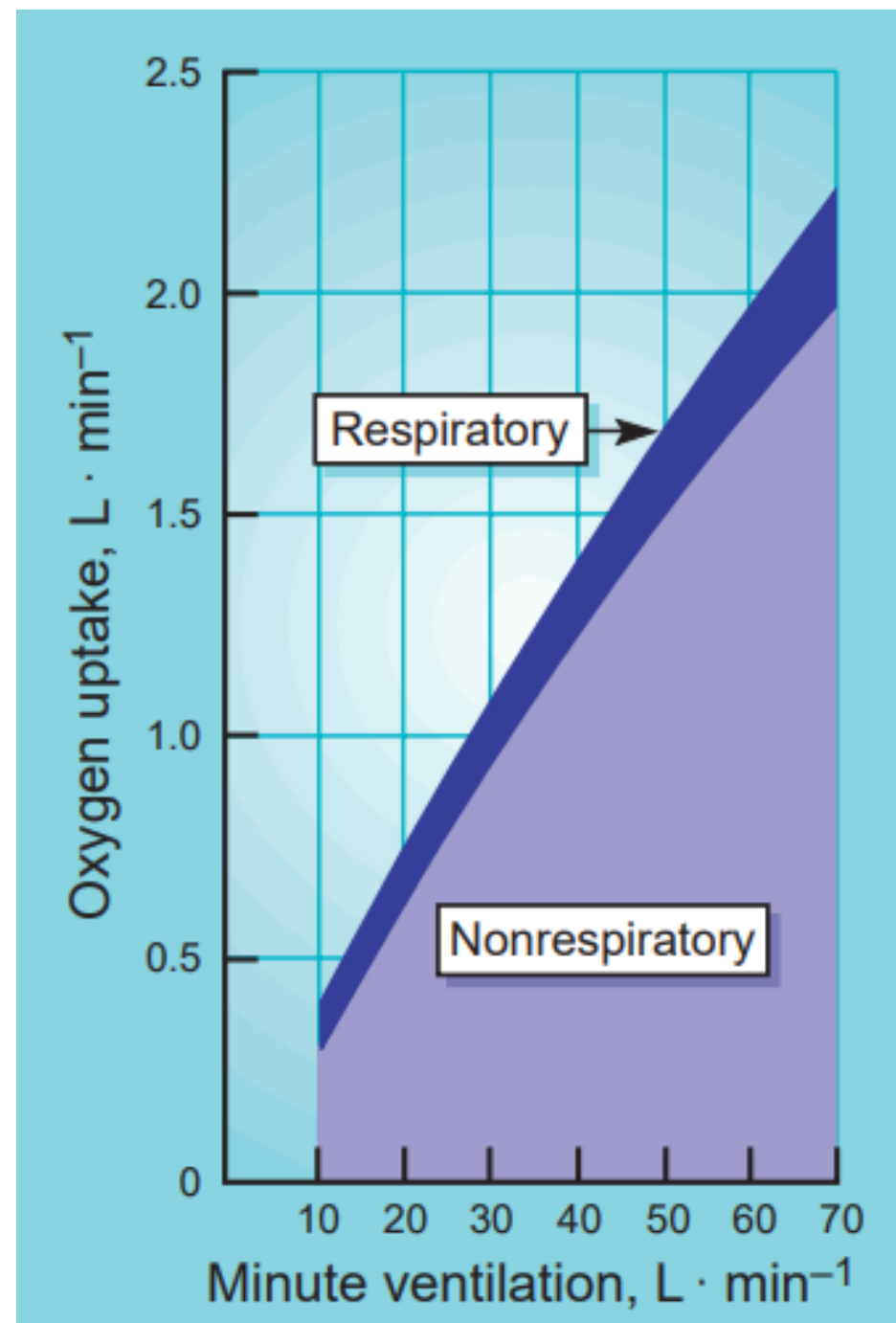
- imply the imbalance between rate of blood lactate clearance/production

Energy requirement of breathing

- compliance of the lung and thorax

- resistance of airways to the air flow

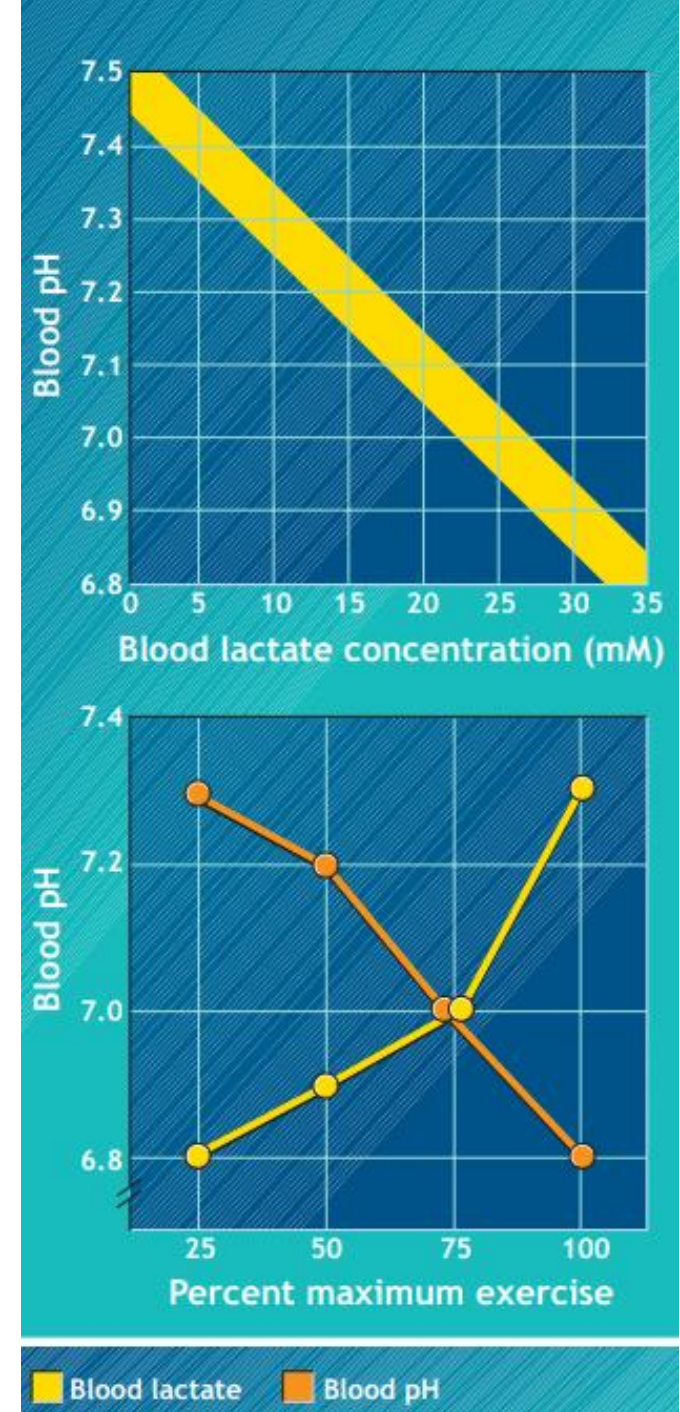
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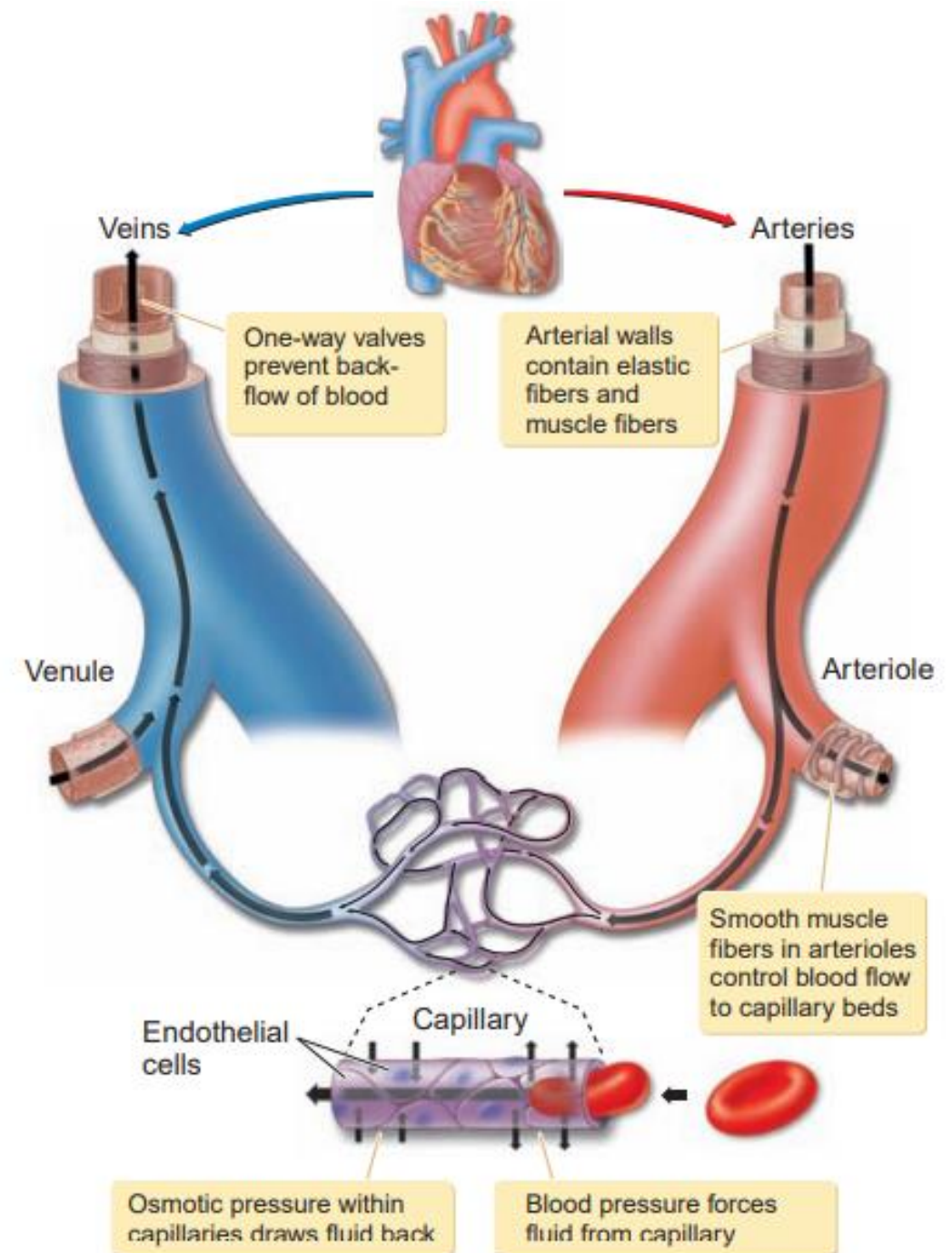
Relationship between blood pH and blood lactate conc

Relationship between blood pH and blood lactate conc related to exercise intensity

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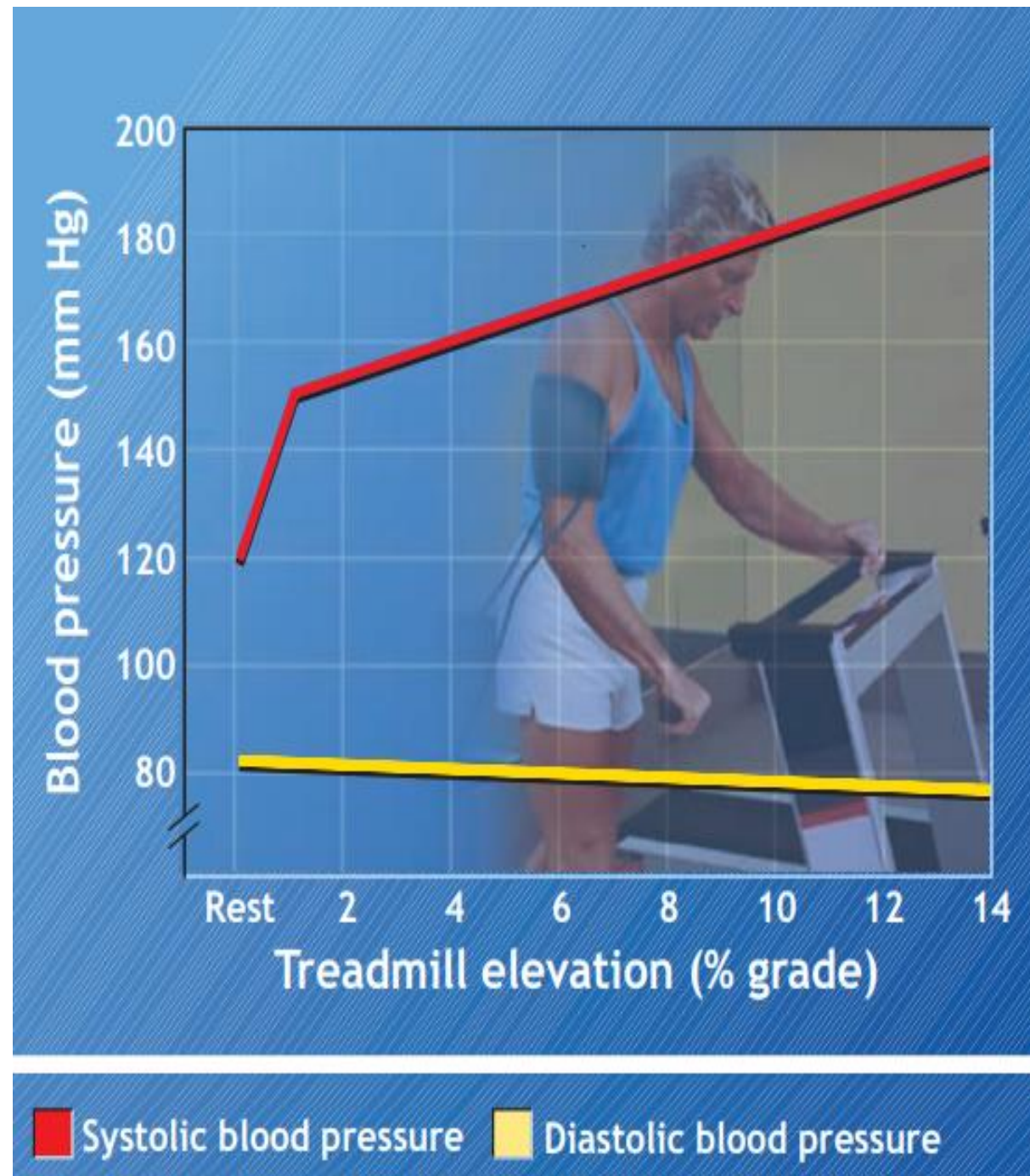
The cardiovascular system and exercise



Rhythmic exercise

- vascular dilate in active muscles
- systolic pressure in the first few min and levels off
- diastolic pressure relatively unchanged
- ↑ pulse pressure

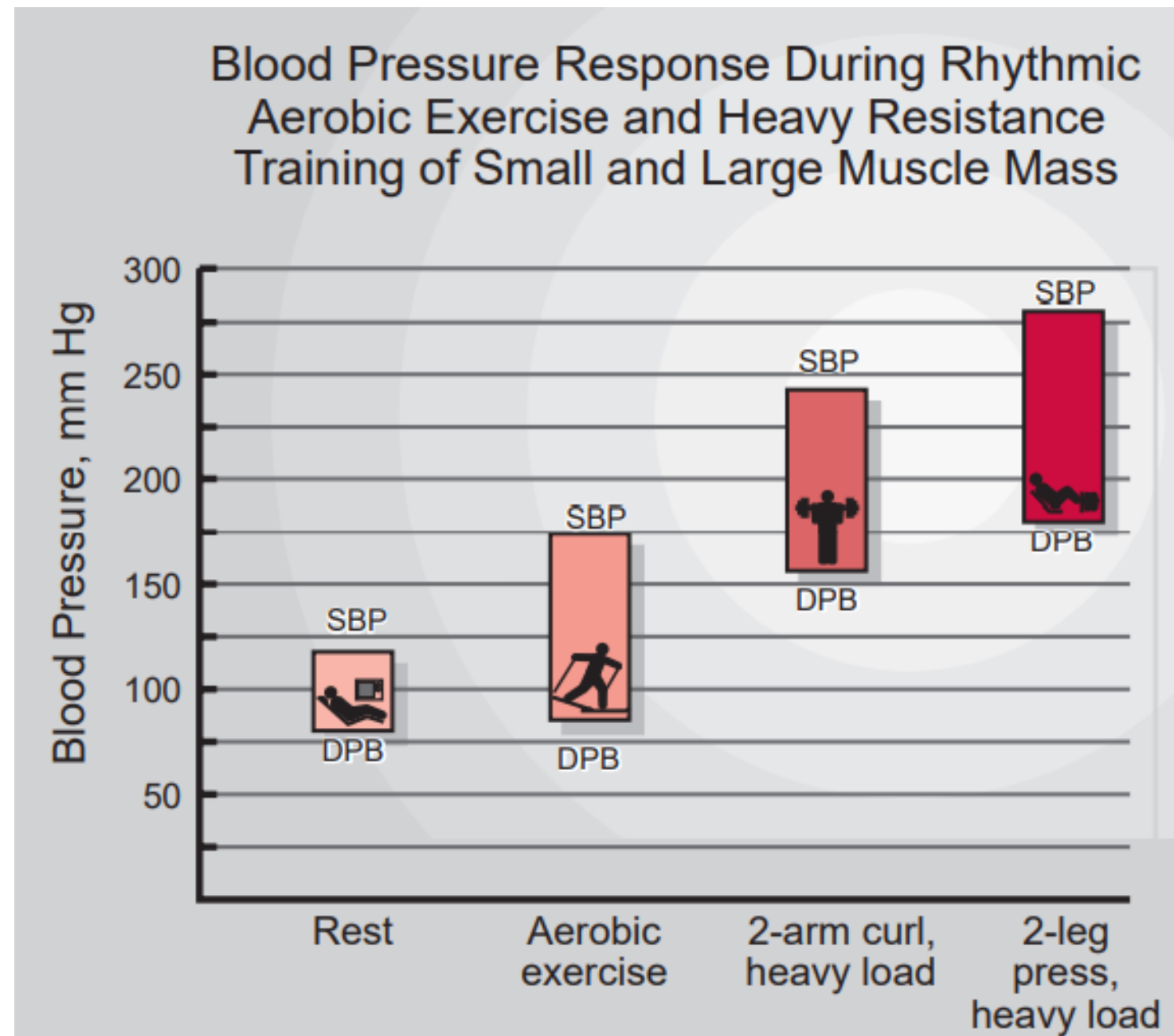
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Resistance exercise

- contrast to rhythmic aerobic exercise
- intense resistance exercise (with muscle mass)
- peripheral artery compressed
- ↑↑↑blood pressure
- increase risk for hypertension

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Regulation of heart rate control

-cortex...medulla....cardio center

-symp/parasymp

-adrenal medulla.....epinephrine

-baroreceptor reflex

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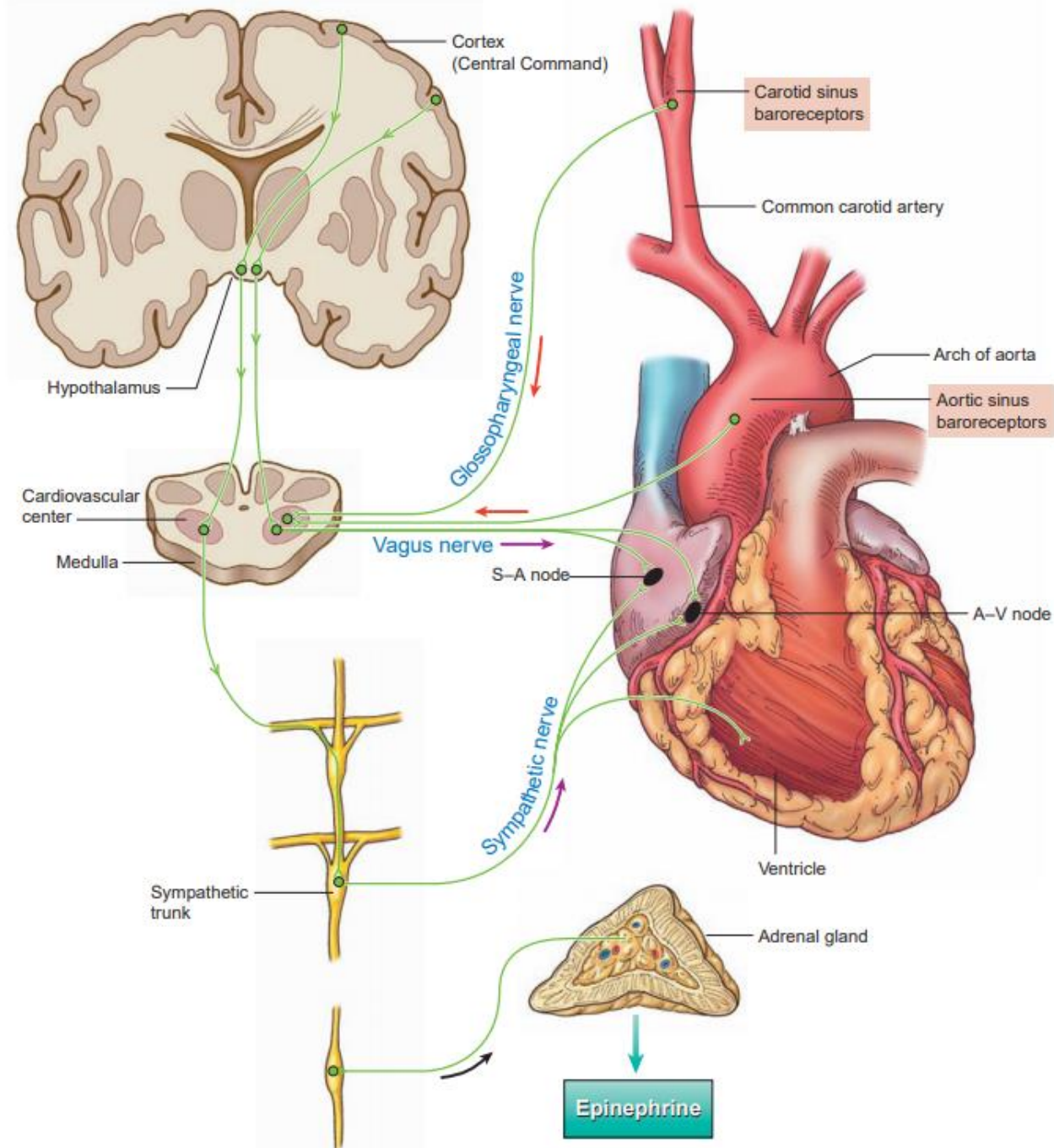


Table 10.2**Summary of Integrated Chemical, Neural, and Hormonal Adjustments Before and During Exercise**

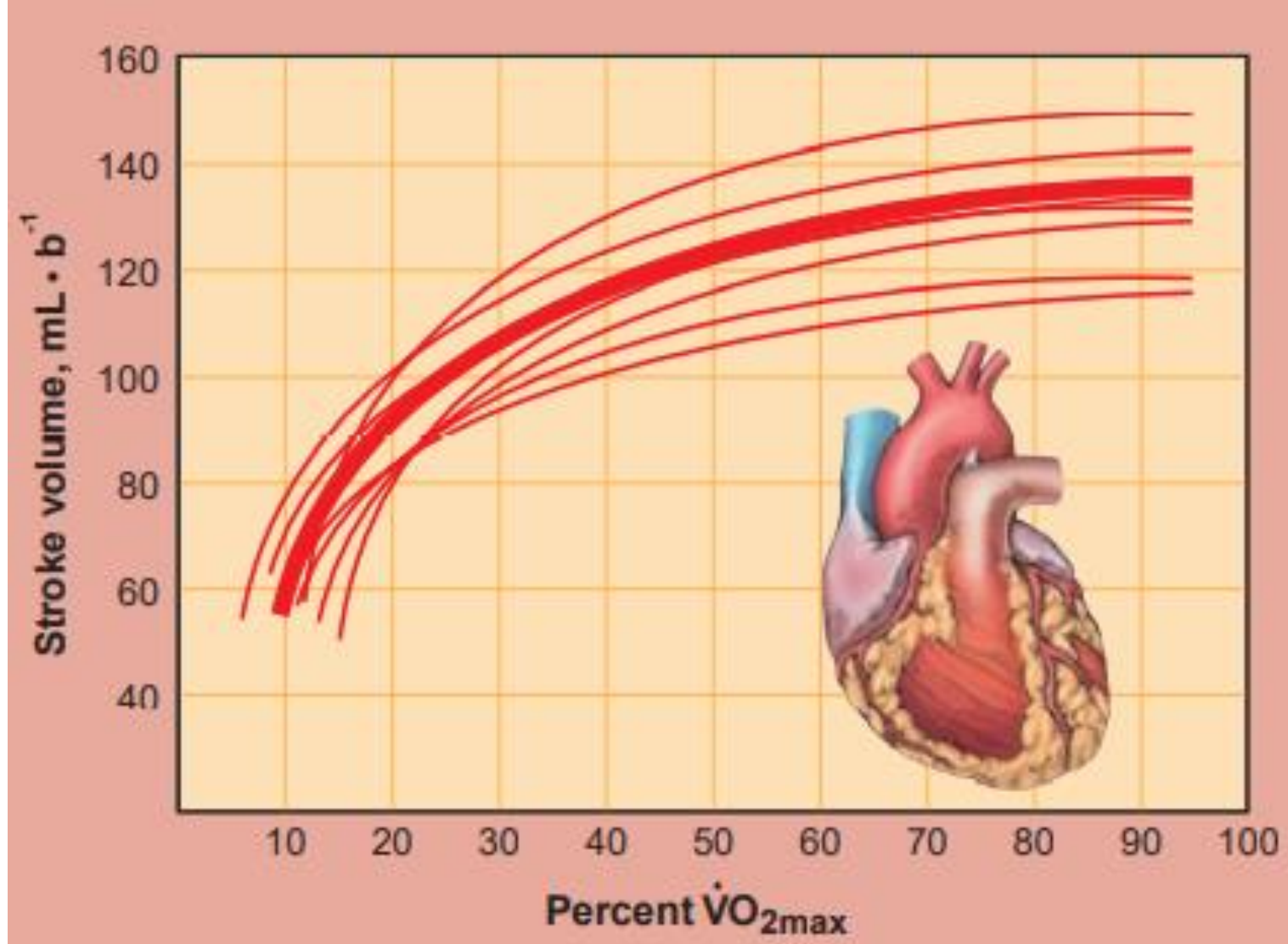
CONDITION	ACTIVATOR	RESPONSE
Pre-exercise “anticipatory” response	Activation of motor cortex and higher areas of brain causes increase in sympathetic outflow and reciprocal inhibition of parasympathetic activity	Acceleration of heart rate; increased myocardial contractility; vasodilation in skeletal and heart muscle (cholinergic fibers); vasoconstriction in other areas, especially skin, gut, spleen, liver, and kidneys (adrenergic fibers); increase in arterial blood pressure
Exercise	Continued sympathetic cholinergic outflow; alterations in local metabolic conditions due to hypoxia ($\downarrow$ pH, $\uparrow$ PCO ₂ , $\uparrow$ ADP, $\uparrow$ Mg ⁺⁺ , $\uparrow$ Ca ⁺⁺ , $\uparrow$ NO, $\uparrow$ temperature) Continued sympathetic adrenergic outflow in conjunction with epinephrine and norepinephrine from the adrenal medulla	Further dilation of muscle vasculature Concomitant constriction of vasculature in inactive tissues to maintain, adequate perfusion pressure throughout the arterial system Venous vessels stiffen to reduce their capacity Venoconstriction facilitates venous return and maintains the central blood volume

At rest

	Cardiac Output Rate (mL · min ⁻¹)	Heart Stroke (b · min ⁻¹)	Volume (mL · b ⁻¹)
Untrained	5000	70	71
Trained	5000	50	100

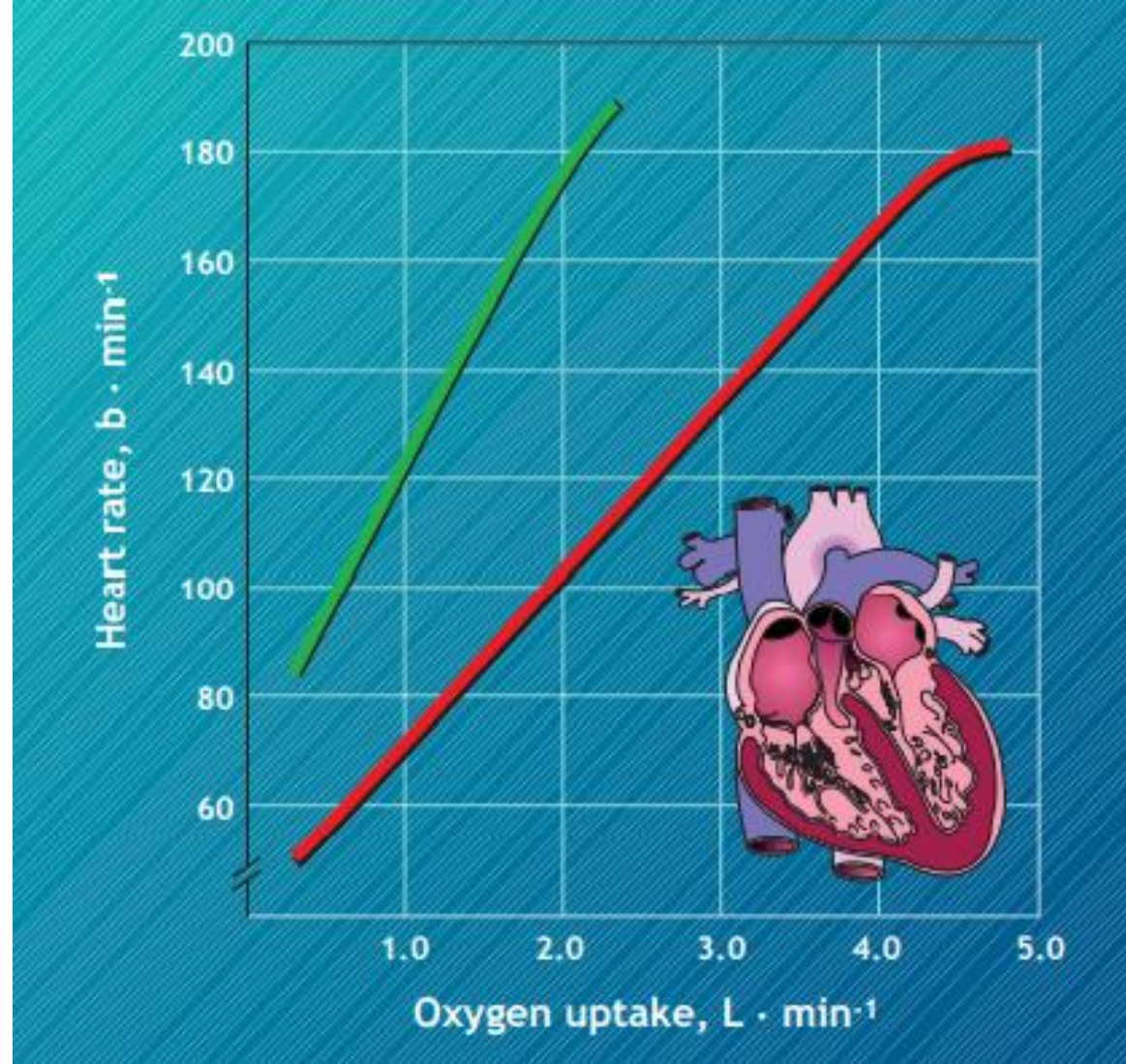
exercise

	Cardiac Output Rate (mL · min ⁻¹)	Heart Stroke (b · min ⁻¹)	Volume (mL · b ⁻¹)
Untrained	22,000	195	113
Trained	35,000	195	179



Stroke volume related to increasing exercise intensity

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Comparison between heart rate of sedentary and endurance-trained subjects related with oxygen uptake

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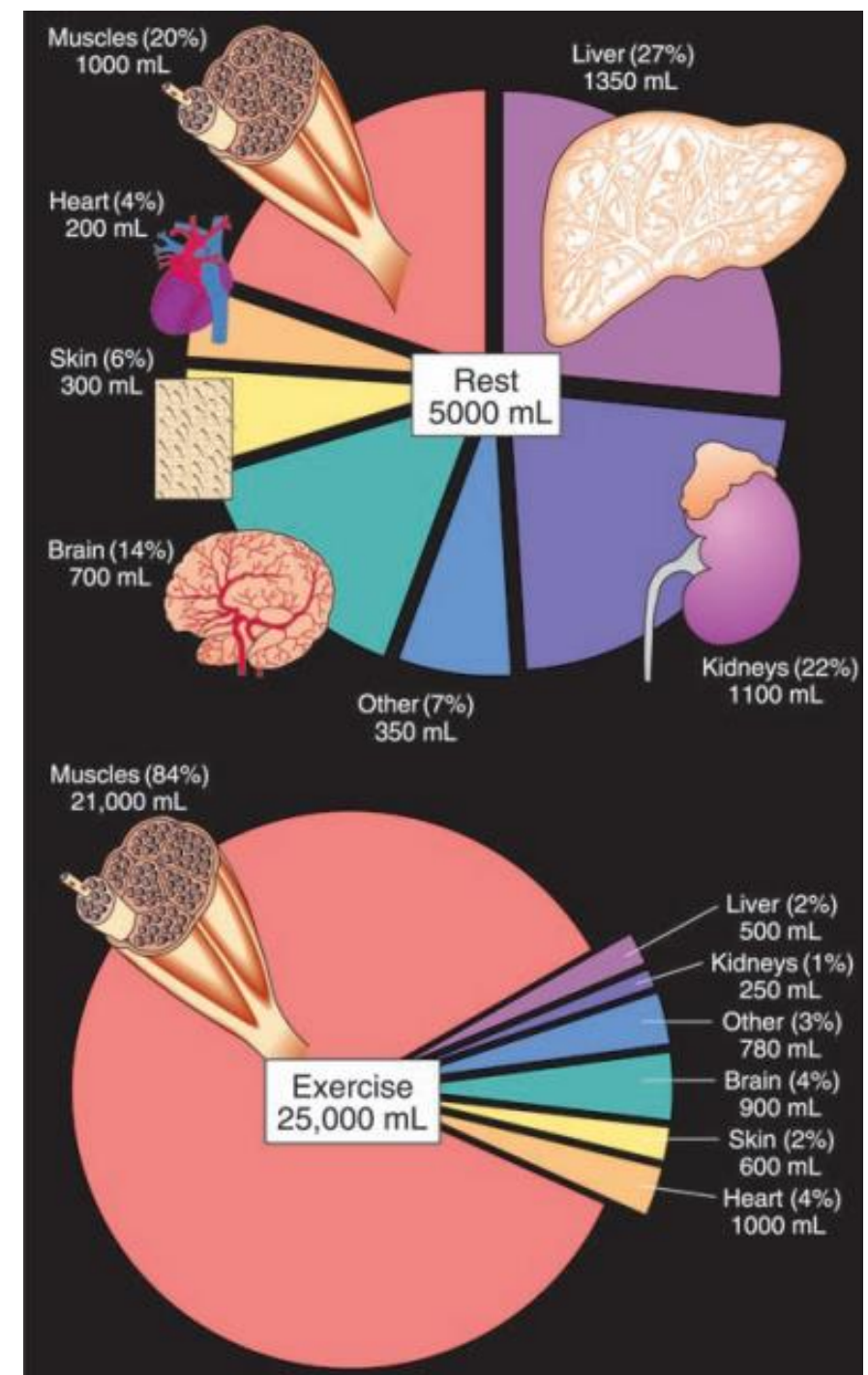
Cardiac output distribution

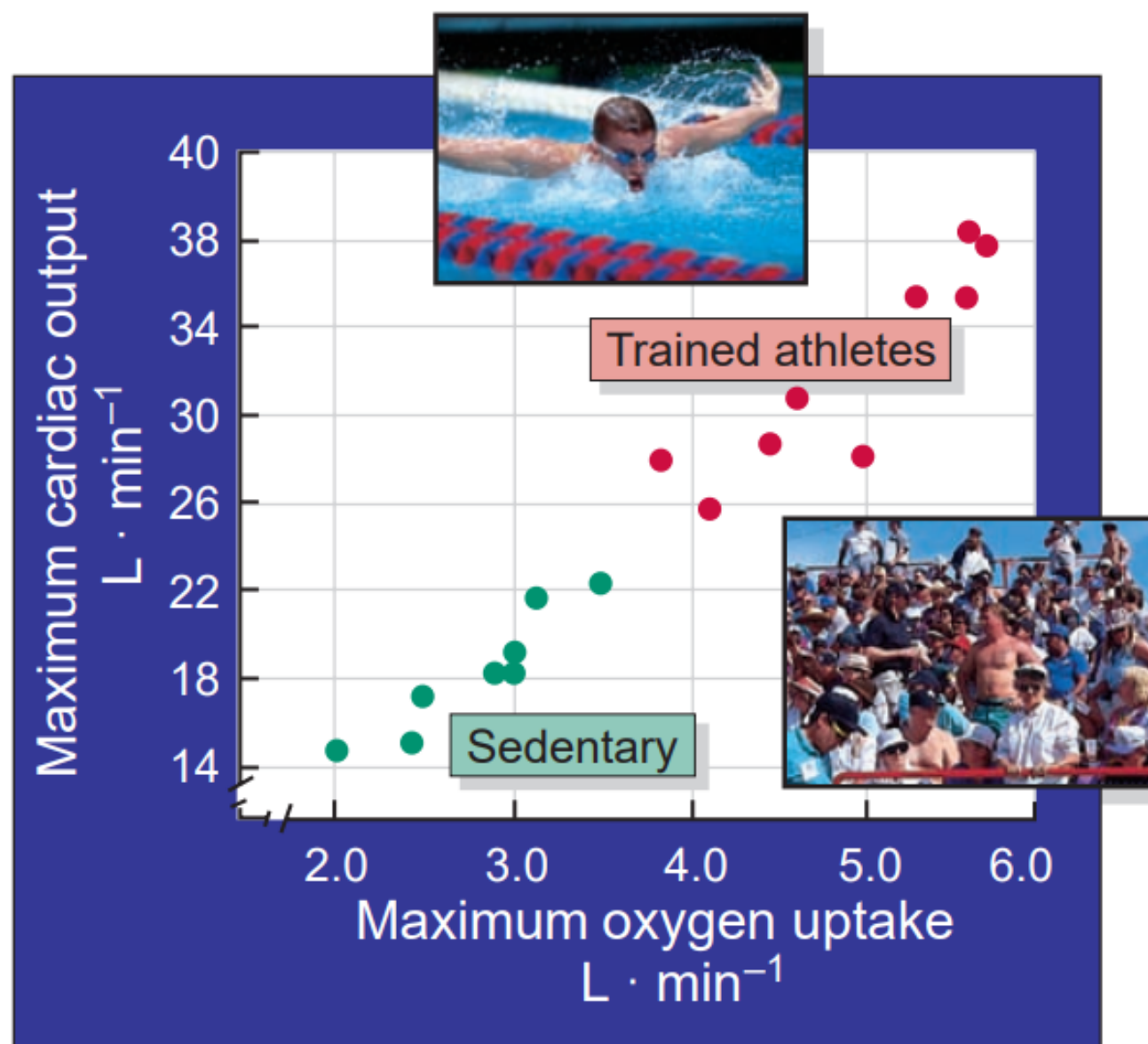
At rest

- more than 1/4 to liver
- more than 1/5 to kidney and muscle

During intense aerobic exercise

- muscle blood flow increases to max at 50 to 75 mL per 100 g of active muscle





Relationship between max cardiac output and max oxygen uptake

Extraction of oxygen

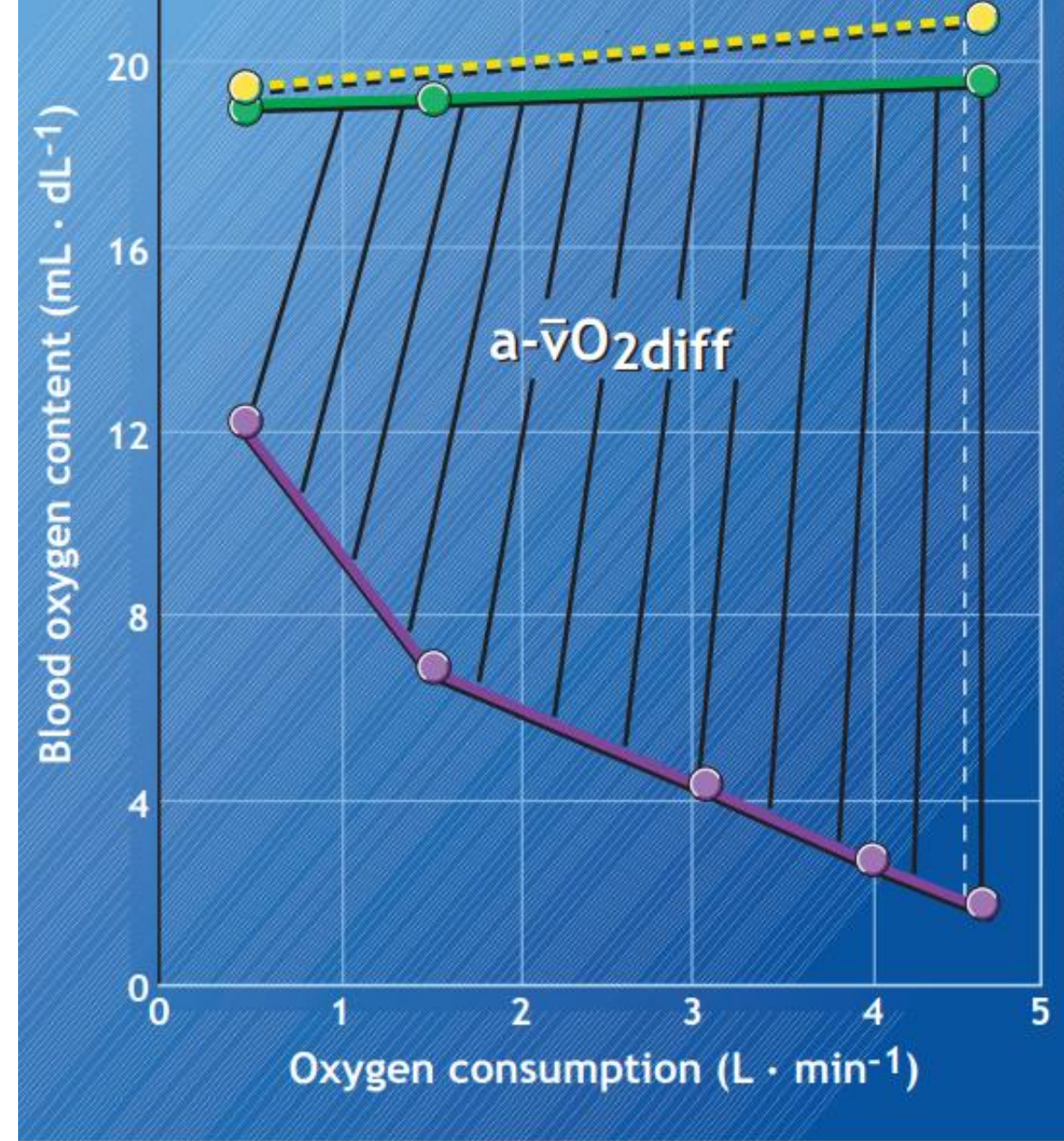
During max exercise

- 20-fold O₂ needed
- blood flow can not match the need
- however, hemoglobin releases extra O₂

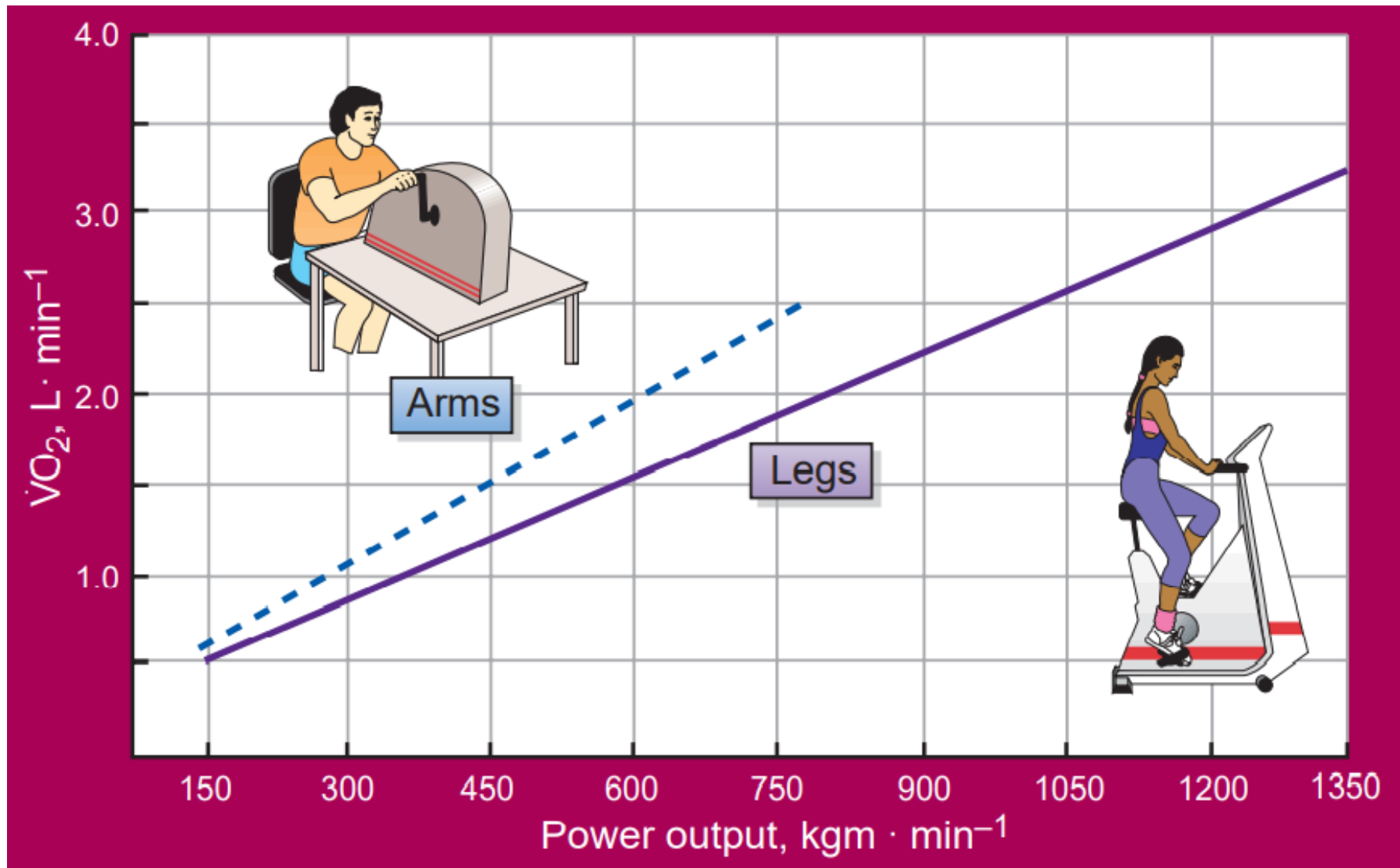
2 mechanisms

- increased tissue blood flow
- use O₂ that is normally unused at rest (the $a-\bar{v}O_2\text{diff}$)

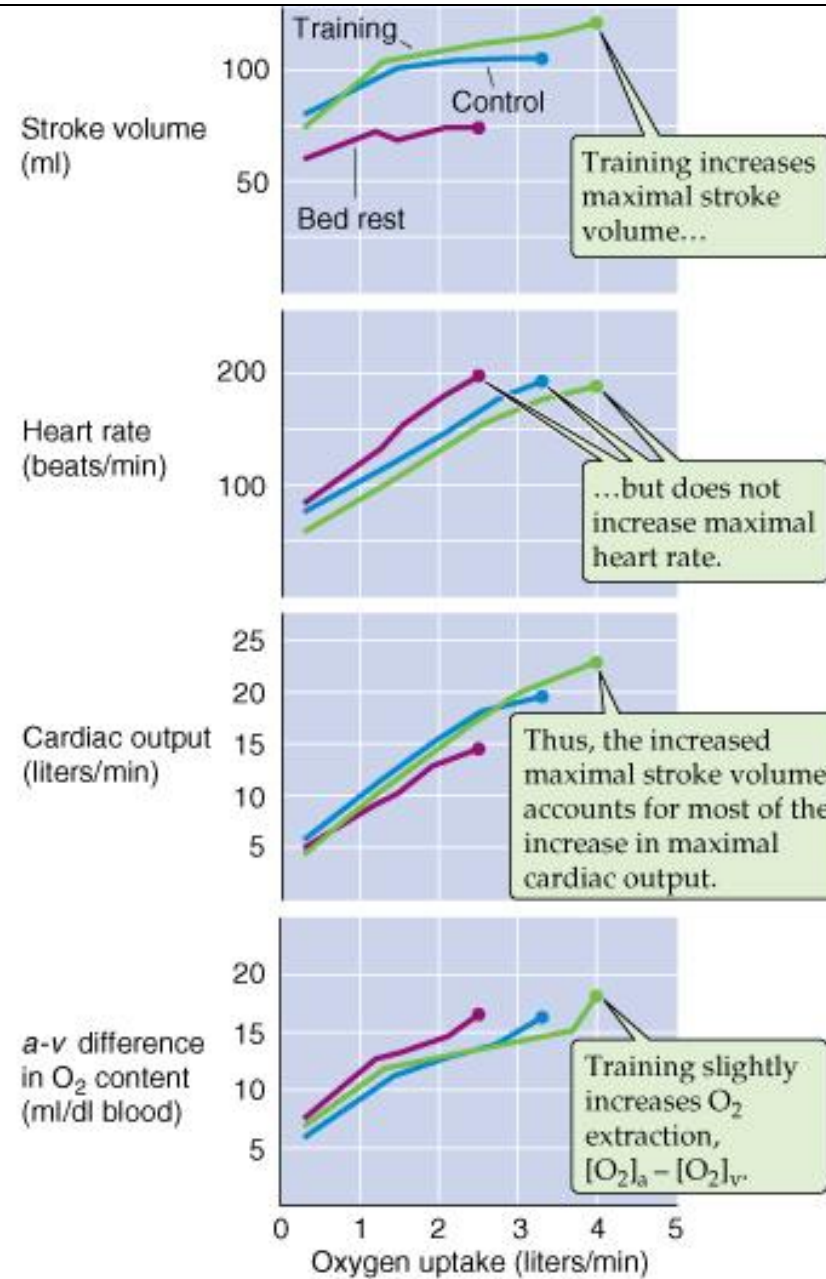
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Arterial O₂ capacity Arterial O₂ content
Mixed-venous O₂ content



Upper body exercise



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Effect of training on various cardiovascular parameters in summary.